

ENERGY RESOURCES



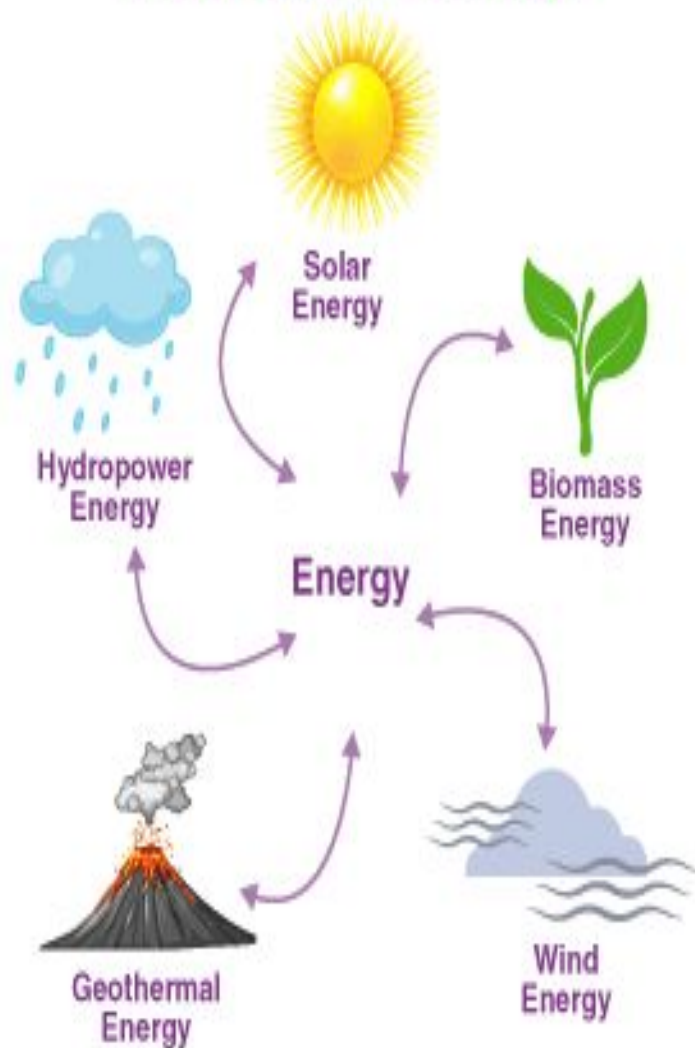
- The sun is the main source of energy on Earth. Other energy sources include coal, geothermal energy, wind energy, biomass, petrol, nuclear energy, and many more. Energy is classified into various types based on sustainability as renewable sources of energy and non-renewable sources of energy.

Sources Of Energy

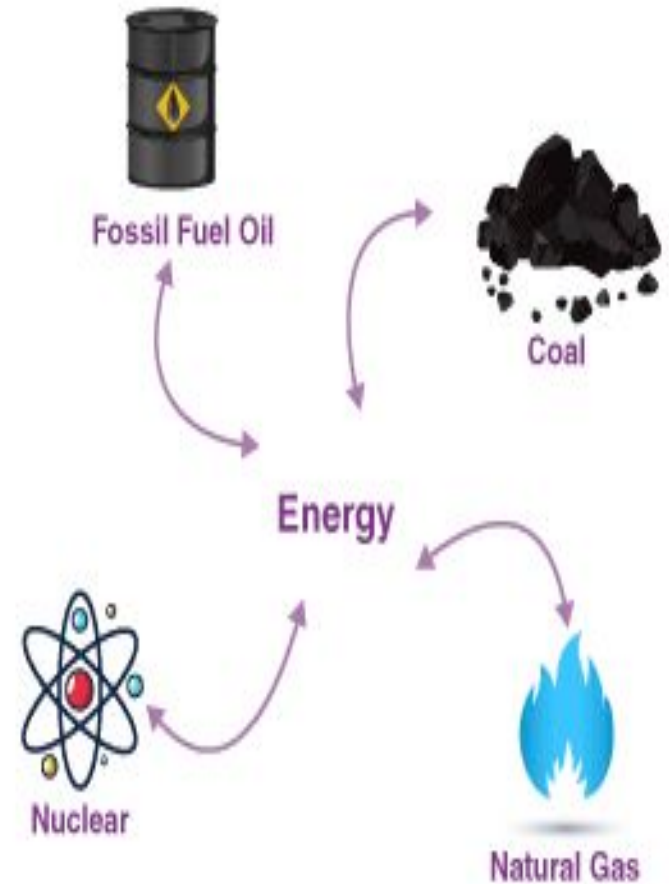


- Sources of energy can be classified into:
- Renewable Sources
- Non-renewable Sources
- Renewable sources of energy are available plentiful in nature and are sustainable. These resources of energy can be naturally replenished and are safe for the environment.
- **Examples of renewable sources of energy are:** Solar energy, geothermal energy, wind energy, biomass, hydropower and tidal energy.
- A [non-renewable resource](#) is a natural resource that is found underneath the earth. These type of energy resources do not replenish at the same speed at which it is used. They take millions of years to replenish. The main examples of non-renewable resources are coal, oil and natural gas.
- **Examples of non-renewable sources of energy are:** Natural gas, coal, petroleum, nuclear energy and hydrocarbon gas liquids.

Renewable Energy



Non-renewable Energy



Difference



Renewable

- The resources that can be renewed once they are consumed are called renewable sources of energy.
- These resources do not cause any environmental pollution.
- Renewable resources are inexhaustible.
- Renewable resources are not affected by human activities.

Non-renewable

- The resources that cannot be renewed once they are consumed are called non-renewable sources of energy.
- These resources cause environmental pollution.
- Non- Renewable resources are exhaustible.
- Non- Renewable resources are affected by human activities.

Solar Energy

- Solar energy, radiation from the Sun capable of producing heat, causing chemical reactions, or generating electricity. The total amount of solar energy incident on Earth is vastly in excess of the world's current and anticipated energy requirements. If suitably harnessed, this highly diffused source has the potential to satisfy all future energy needs. In the 21st century solar energy is expected to become increasingly attractive as a renewable energy source because of its inexhaustible supply and its nonpolluting character, in stark contrast to the finite fossil fuels coal, petroleum, and natural gas.

● **APPLICATIONS OF SOLAR ENERGY**

- Some of the major application of solar energy are as follows: (a) Solar water heating (b) Solar heating of buildings (c) Solar distillation (d) Solar pumping (e) Solar drying of agricultural and animal products (f) Solar furnaces (g) Solar cooking (h) Solar electric power generation (i) Solar thermal power production
- **(a) Solar Water Heating:**
- A solar water heating unit comprises a blackened flat plate metal collector with an associated metal tubing facing the general direction of the sun. The plate collector has a transparent glass cover above and a layer of thermal insulation beneath it.
- The metal tubing of the collector is connected by a pipe to an insulated tank that stores hot water during cloudy days. The collector absorbs solar radiations and transfers the heat to the water circulating through the tubing either by gravity or by a pump.
- This hot water is supplied to the storage tank via the associated metal tubing. This system of water heating is commonly used in hotels, guest houses, tourist bungalows, hospitals, canteens as well as domestic and industrial units.

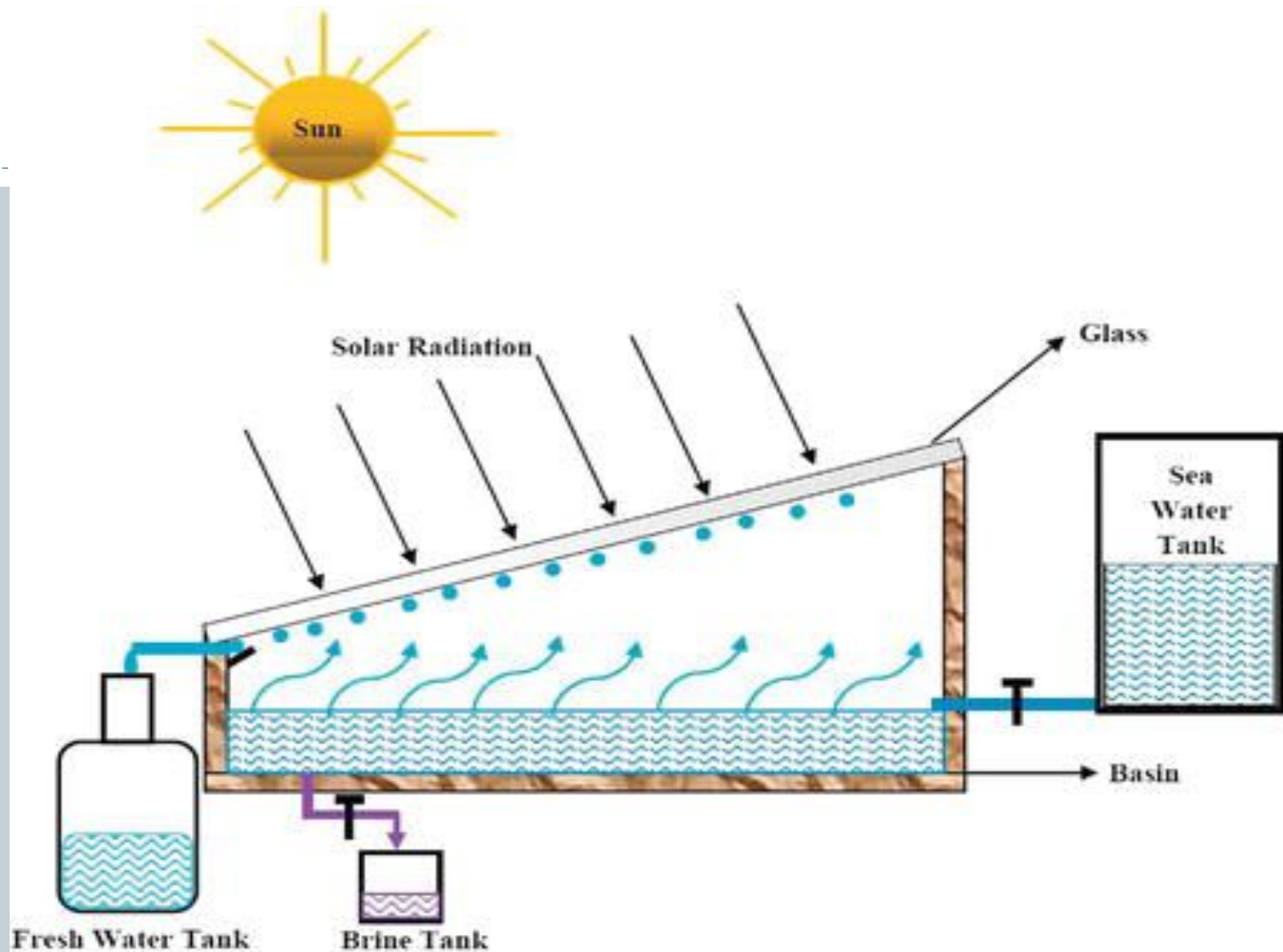


- **(b) Solar Heating of Buildings:**

- Using separate solar collectors which may heat either water or air or storage devices which can accumulate the collected solar energy for use at night and during inclement days.
- When the building requires heat then from these collectors or storage devices, the heat is transferred by conventional equipment such as fan, ducts, air outlets, radiators and hot air registers etc. to warm up the living spaces of a building.
- When the building does not require heat, the heated air or water from the collector can be moved to the heat storage device such as well insulated water tank or other heat holding material. For inclement days, an auxiliary heating system using gas, oil or electricity is required as a backup system.

- **(c) Solar-distillation:**

- In arid semi and or coastal areas there is scarcity of potable water. The abundant sunlight in these areas can be used for converting saline water into potable distilled water by the method of solar distillation. In this method, solar radiation is admitted through a transparent air tight glass cover into a shallow blackened basin containing saline water.
- Solar radiation passes through the covers and is absorbed and converted into heat in the blackened surface causing the water to evaporate from the brine (impure saline water). The vapors produced get condensed to form purified water in the cool interior of the roof.
- The condensed water flows down the sloping roof and is collected in the troughs placed at the bottom and from there into a water storage tank to supply potable distilled water in areas of scarcity



- **(d) Solar-pumping:**

- In solar pumping, the power generated by solar-energy is utilized for pumping water for irrigation purposes. The requirement for water pumping is greatest in the hot summer months which coincide with the increased solar radiations during this period and so this method is most appropriate for irrigation purpose. During periods of inclement weather when solar radiations are low then the requirement for water pumping is also relatively less as the transpiration losses from the crops are also low.





- **(h) Solar Electric Power Generation:**

- Electric energy or electricity can be produced directly from solar energy by means of photovoltaic cells. The photovoltaic cell is an energy conversion device which is used to convert photons of sunlight directly into electricity. It is made of semi conductors which absorb the photons received from the sun, creating free electrons with high energies.
- These high energy free electrons are induced by an electric field, to flow out of the semiconductor to do useful work. This electric field in photovoltaic cells is usually provided by a p-n junction of materials which have different electrical properties. There are different fabrication techniques to enable these cells to achieve maximum efficiency.
- These cells are arranged in parallel or series combination to form cell modules. Some of the special features of these modules are high reliability, no expenditure on fuel, minimum cost of maintenance, long life, portability, modularity, pollution free working etc.
- Photovoltaic cells have been used to operate irrigation pumps, rail road crossing warnings, navigational signals, highway emergency call systems, automatic meteorological stations etc. in areas where it is difficult to lay power lines.
- They are also used for weather monitoring and as portable power sources for televisions, calculators, watches, computer card readers, battery charging and in satellites etc. Besides these, photovoltaic cells are used for the energisation of pump sets for irrigation, drinking water supply and for providing electricity in rural areas i.e. street lights etc.

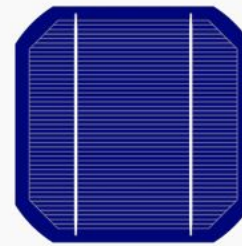
● (i) Solar Thermal Power Production:

- Solar thermal power production means the conversion of solar energy into electricity through thermal energy. In this procedure, solar energy is first utilized to heat up a working fluid, gas, water or any other volatile liquid. This heat energy is then converted into mechanical energy by a turbine. Finally a conventional generator coupled to a turbine converts this mechanical energy into electrical energy.

● Production of Power through Solar Ponds:

- A solar pond is a natural or artificial body of water utilized for collecting and absorbing solar radiation and storing it as heat. It is very shallow (5-10 cm deep) and has a radiation absorbing (black plastic) bottom. It has a curved fiber glass cover over it to permit the entry of solar radiation but reduces losses by radiation and convection (air movement). Loss of heat to the ground is minimized by providing a bed of insulating material under the pond.
- Solar ponds utilize water for collecting and storing the solar energy which is used for many applications such as space heating, industrial process heating and to generate electricity by driving a turbine powered by evaporating an organic fluid with a low boiling point.

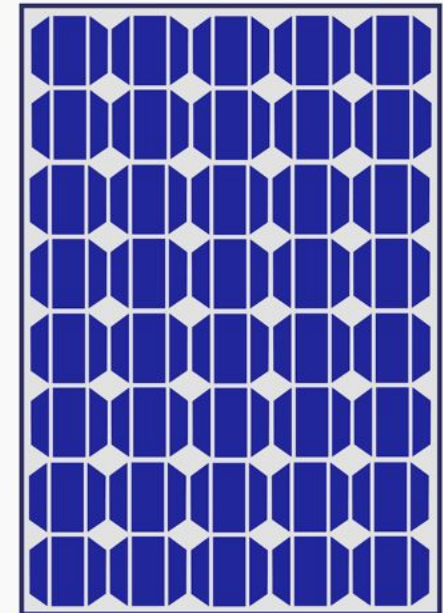
From a solar cell to a PV System



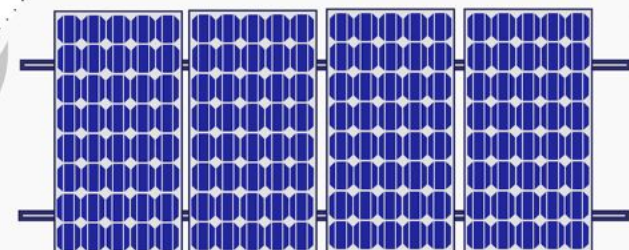
Solar Cell



Solar Module



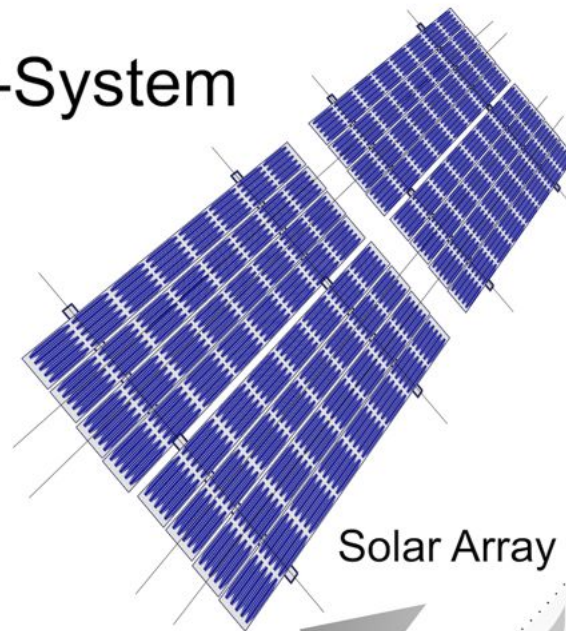
Solar Panel



Solar Array



PV-System



Electricity Meter

AC Isolator

Fusebox

Inverter

Battery

Charge Controller

Generation Meter

DC Isolator

Cabling

Mounting

Tracking System

Wind Energy

- **Wind energy**, form of solar energy that is produced by the movement of air relative to Earth's surface. This form of energy is generated by the uneven heating of Earth's surface by the Sun and is modified by Earth's rotation and surface topography. .
- Wind energy is a form of renewable energy, typically powered by the movement of wind across enormous fan-shaped structures called wind turbines. Once built, these turbines create no climate-warming greenhouse gas emissions, making this a “carbon-free” energy source that can provide electricity without making climate change worse. Wind energy is the third-largest source of carbon-free electricity in the world (after hydropower and nuclear) and the second-fastest-growing (after solar).

Hydroelectricity

- **Hydroelectric power**, electricity produced from generators driven by turbines that convert the potential energy of falling or fast-flowing water into mechanical energy. In the early 21st century, hydroelectric power was the most widely utilized form of renewable energy; in 2019 it accounted for more than 18 percent of the world's total power generation capacity.
- Hydroelectric power plants are usually located in dams that impound rivers, thereby raising the level of the water behind the dam and creating as high a head as is feasible. The potential power that can be derived from a volume of water is directly proportional to the working head, so that a high-head installation requires a smaller volume of water than a low-head installation to produce an equal amount of power. In some dams, the powerhouse is constructed on one flank of the dam, part of the dam being used as a spillway over which excess water is discharged in times of flood. Where the river flows in a narrow steep gorge, the powerhouse may be located within the dam itself.

Tidal And Wave Power Energy



- Tides are a regular phenomenon. They can be predicted over months and years in advance. This is why the energy from this massive movement of water can be harnessed and converted into a usable form of energy.
- Wave Energy is a type of renewable energy that can be obtained by using the motion of waves. Waves are created when the wind blows across the ocean's surface. The amount of energy transferred and the magnitude of the subsequent waves are governed by the wind speed, the duration for which the wind blows, and the distance the wind blows over.

NON RENEWABLE ENERGY SOURCES

- **Types of Non-renewable Energy**

- Non-renewable energy primarily consists of fossil fuels and nuclear fuels. Let's explore each of these types in detail:



- **Fossil Fuels**

- Fossil fuels are formed from the remains of ancient plants and animals that underwent a process of decomposition over millions of years. They are categorized into three main types:

- **Natural Gas**

- Natural gas is a gaseous fossil fuel composed mainly of methane. It is often found in association with oil deposits. Natural gas is a versatile energy source used for heating, cooking, electricity generation, and as a fuel for vehicles.

- **Coal**

- Coal is a solid fossil fuel that forms when plant material, such as trees, plants, and ferns, undergoes decomposition under high pressure and heat over millions of years. It is primarily used for electricity generation in power plants and is known for its significant carbon content.

- **Oil**

- Oil is a liquid fossil fuel that results from the decomposition of smaller organisms like zooplankton and algae, which were subjected to immense pressure over time. Oil is a crucial energy resource used in transportation, manufacturing, and producing various petrochemical products.

- **Nuclear Fuels**

- Nuclear fuels are utilised in nuclear power plants, where energy is generated through nuclear fission. The most commonly used nuclear fuel is uranium, found in the Earth's crust in relatively low concentrations. Uranium undergoes fission reactions, releasing large amounts of energy.
- Understanding the significance of both fossil fuels and nuclear fuels as non-renewable energy sources is important. While they have played a crucial role in meeting our energy demands, their extraction, use, and environmental impact must be carefully managed.

ADVANTAGES



- Non-renewable sources of energy, such as diesel and oil, are known for their affordability, making them accessible to a wide range of applications.
- Non-renewable energy sources offer ease of accessibility and compatibility with existing infrastructure.
- These energy sources are relatively easy to store, providing convenience in meeting energy demands.

DISADVANTAGES



- Non-renewable energy is finite in nature, once depleted, it cannot be replenished.
- The by-products generated from non-renewable energy production contribute to environmental damage and an increase in greenhouse gas emissions.
- The transportation, mining, and extraction activities associated with non-renewable energy sources can result in accidents such as oil spills, nuclear meltdowns, pipeline leaks, and explosions, causing severe harm to the environment.

NUCLEAR ENERGY



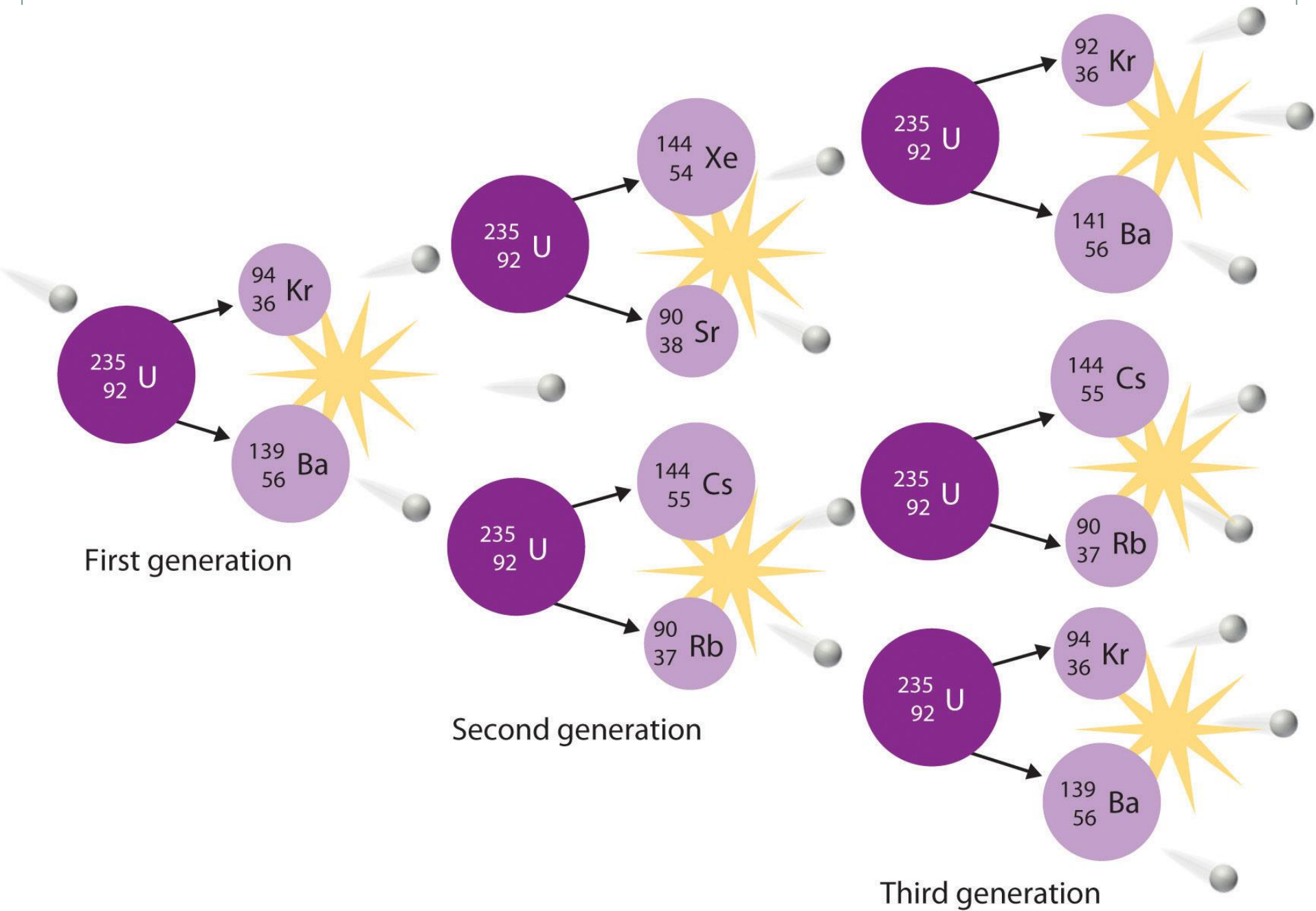
- Nuclear energy is a form of energy released from the nucleus, the core of atoms, made up of protons and neutrons. This source of energy can be produced in two ways: fission – when nuclei of atoms split into several parts – or fusion – when nuclei fuse together.
- The nuclear energy harnessed around the world today to produce electricity is through nuclear fission, while technology to generate electricity from fusion is at the R&D phase. This article will explore nuclear fission.

What is nuclear fission?

Nuclear fission is a reaction where the nucleus of an atom splits into two or more smaller nuclei, while releasing energy.

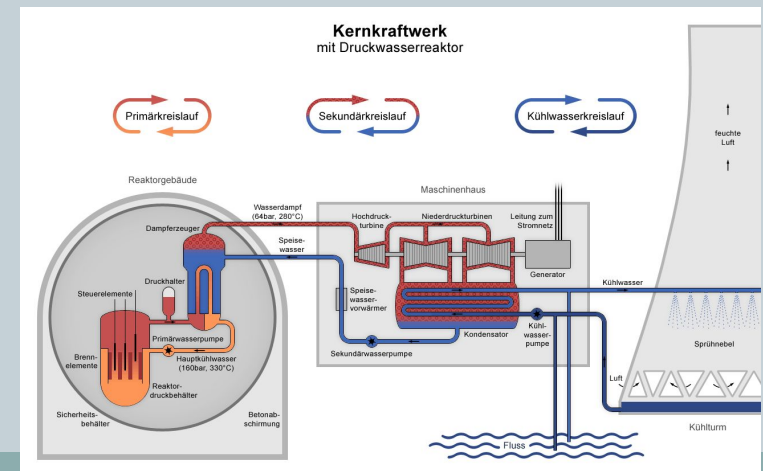
For instance, when hit by a neutron, the nucleus of an atom of uranium-235 splits into two smaller nuclei, for example a barium nucleus and a krypton nucleus and two or three neutrons. These extra neutrons will hit other surrounding uranium-235 atoms, which will also split and generate additional neutrons in a multiplying effect, thus generating a chain reaction in a fraction of a second.

Each time the reaction occurs, there is a release of energy in the form of heat and [radiation](#). The heat can be converted into electricity in a nuclear power plant, similarly to how heat from fossil fuels such as coal, gas and oil is used to generate electricity.



How does a nuclear power plant work?

- Inside nuclear power plants, nuclear reactors and their equipment contain and control the chain reactions, most commonly fuelled by uranium-235, to produce heat through fission. The heat warms the reactor's cooling agent, typically water, to produce steam. The steam is then channelled to spin turbines, activating an electric generator to create low-carbon electricity.



Mining, enrichment and disposal of uranium



- Uranium is a metal that can be found in rocks all over the world. Uranium has several naturally occurring [isotopes](#), which are forms of an element differing in mass and physical properties but with the same chemical properties. Uranium has two primordial isotopes: uranium-238 and uranium-235. Uranium-238 makes up the majority of the uranium in the world but cannot produce a fission chain reaction, while uranium-235 can be used to produce energy by fission but constitutes less than 1 per cent of the world's uranium.
- To make natural uranium more likely to undergo fission, it is necessary to increase the amount of uranium-235 in a given sample through a process called uranium enrichment. Once the uranium is enriched, it can be used effectively as nuclear fuel in power plants for three to five years, after which it is still radioactive and has to be disposed of following stringent guidelines to protect people and the environment. Used fuel, also referred to as spent fuel, can also be recycled into other types of fuel for use as new fuel in special nuclear power plants.

Nuclear power and climate change



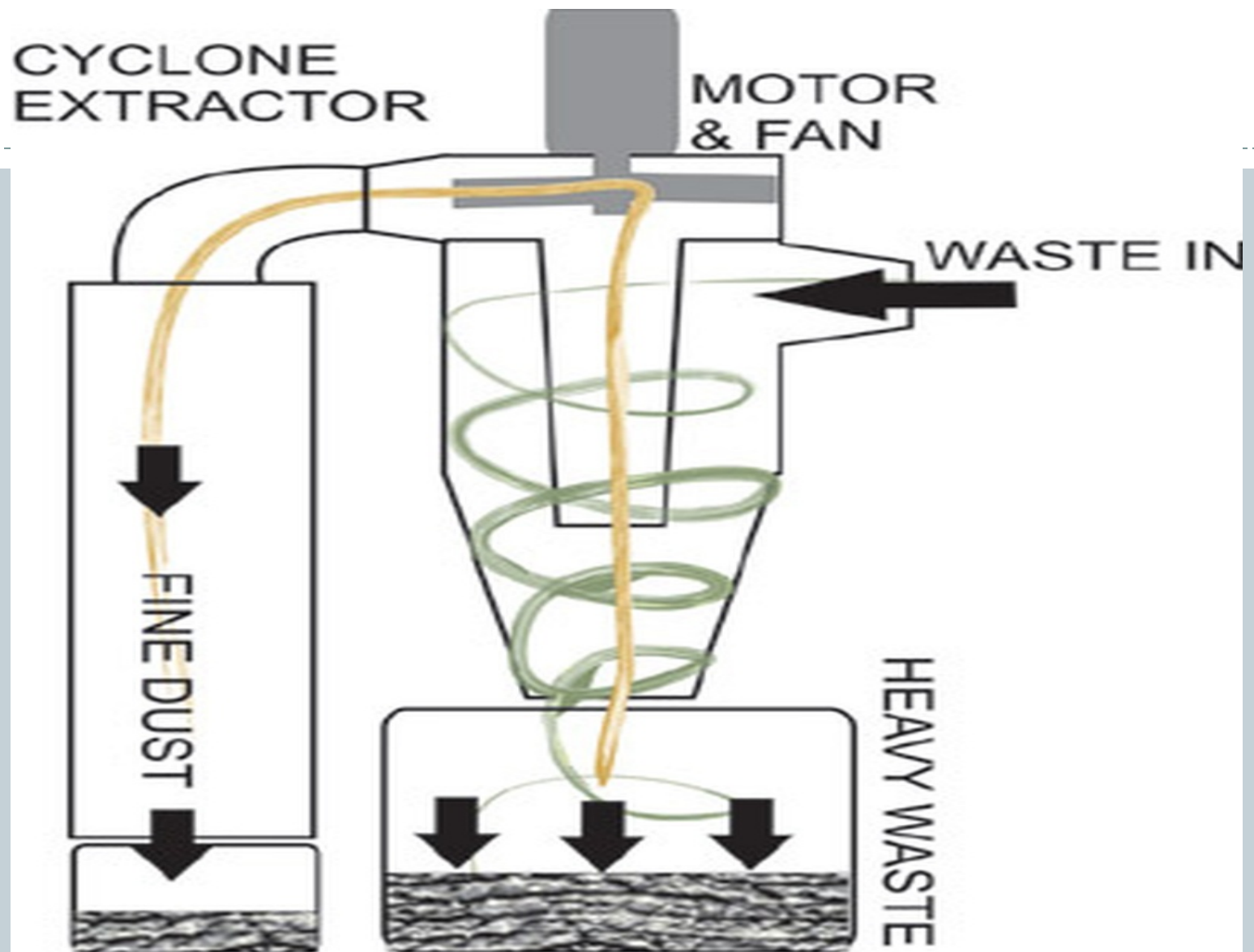
- Nuclear power is a low-carbon source of energy, because unlike coal, oil or gas power plants, nuclear power plants practically do not produce CO₂ during their operation. Nuclear reactors generate close to one-third of the world's carbon free electricity and are crucial in meeting climate change goals.



- **Air pollution control**, the techniques employed to reduce or eliminate the emission into the atmosphere of substances that can harm the environment or human health. The control of air pollution is one of the principal areas of pollution control, along with wastewater treatment, solid-waste management, and hazardous-waste management

● Cyclones

- A cyclone removes particulates by causing the dirty airstream to flow in a spiral path inside a cylindrical chamber. Dirty air enters the chamber from a tangential direction at the outer wall of the device, forming a vortex as it swirls within the chamber. The larger particulates, because of their greater inertia, move outward and are forced against the chamber wall. Slowed by friction with the wall surface, they then slide down the wall into a conical dust hopper at the bottom of the cyclone. The cleaned air swirls upward in a narrower spiral through an inner cylinder and emerges from an outlet at the top. Accumulated particulate dust is periodically removed from the hopper for disposal.
- Cyclones are best at removing relatively coarse particulates. They can routinely achieve efficiencies of 90 percent for particles larger than about 20 micrometres (μm ; 20 millionths of a metre). By themselves, however, cyclones are not sufficient to meet stringent air quality standards. They are typically used as pre-cleaners and are followed by more efficient air-cleaning equipment such as electrostatic precipitators and baghouses (described below).



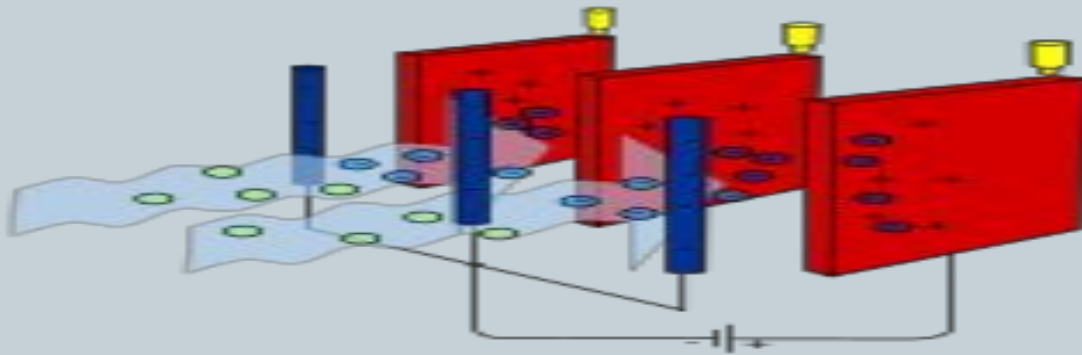
- **Scrubbers**

- Devices called wet scrubbers trap suspended particles by direct contact with a spray of water or other liquid. In effect, a scrubber washes the particulates out of the dirty airstream as they collide with and are entrained by the countless tiny droplets in the spray.
- Several configurations of wet scrubbers are in use. In a spray-tower scrubber, an upward-flowing airstream is washed by water sprayed downward from a series of nozzles. The water is recirculated after it is sufficiently cleaned to prevent clogging of the nozzles. Spray-tower scrubbers can remove 90 percent of particulates larger than about 8 μm .

- **Electrostatic precipitators**

- electrostatic precipitator Electrostatic precipitator, a common particle-collection device at fossil-fuel power-generating stations.(more)

- Electrostatic precipitation is a commonly used method for removing fine particulates from airstreams. In an electrostatic precipitator, particles suspended in the airstream are given an electric charge as they enter the unit and are then removed by the influence of an electric field. The precipitation unit comprises baffles for distributing airflow, discharge and collection electrodes, a dust clean-out system, and collection hoppers. A high voltage of direct current (DC), as much as 100,000 volts, is applied to the discharge electrodes to charge the particles, which then are attracted to oppositely charged collection electrodes, on which they become trapped.



- In a typical unit the collection electrodes comprise a group of large rectangular metal plates suspended vertically and parallel to each other inside a boxlike structure. There are often hundreds of plates having a combined surface area of tens of thousands of square metres. Rows of discharge electrode wires hang between the collection plates. The wires are given a negative electric charge, whereas the plates are grounded and thus become positively charged.
- Particles that stick to the collection plates are removed periodically when the plates are shaken, or “rapped.” Rapping is a mechanical technique for separating the trapped particles from the plates, which typically become covered with a 6-mm (0.2-inch) layer of dust. Rappers are either of the impulse (single-blow) or vibrating type. The dislodged particles are collected in a hopper at the bottom of the unit and removed for disposal. An electrostatic precipitator can remove particulates as small as 1 μm with an efficiency exceeding 99 percent. The effectiveness of electrostatic precipitators in removing fly ash from the combustion gases of fossil-fuel furnaces accounts for their high frequency of use at power stations.

TEMPERATURE INVERSION



- **Temperature inversion**, a reversal of the normal behaviour of temperature in the troposphere (the region of the atmosphere nearest Earth's surface), in which a layer of cool air at the surface is overlain by a layer of warmer air. (Under normal conditions air temperature usually decreases with height.)
- Inversions play an important role in determining cloud forms, precipitation, and visibility. An inversion acts as a cap on the upward movement of air from the layers below. As a result, convection produced by the heating of air from below is limited to levels below the inversion. Diffusion of dust, smoke, and other air pollutants is likewise limited.